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Parasitic Memcapacitor Effects on HP TiO₂ Memristor Dynamics

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ABSTRACT In this paper, we study an HP TiO₂ memristor model with a parasitic memcapacitor, and it is shown that the parasitic element has significant effects on the volt-ampere characteristics and the dynamics of the memristor circuits. Further, the study shows that the pinched point on the hysteresis loop of the HP TiO₂ memristor may deviate from the origin. We also show that the memristor with a parasitic memcapacitor will eventually lose its stored flux when its power is switched off. Furthermore, in order to study the parasitic effects in circuits, we derive a simple three-element series circuit that contains an inductor, a negative resistor, and an HP TiO₂ memristor. If the parasitic memcapacitor of the memristor is considered, we observe a parasitic periodic oscillation occurred in the circuit.

INDEX TERMS Memristor, parasitic memcapacitor, periodic oscillation.

I. INTRODUCTION

A memristor is a nonlinear resistor with memory, which was first postulated in 1971 by Chua as the fourth basic circuit element [1], and generalized in 1976 to a broader class of dynamical systems called memristive devices and systems [2]. However, the memristor attracted most of attention and interest only after 2008, when the first practical memristor implementation was reported by the Hewlett-Packard [3]. Other than the HP TiO₂ memristor, some memristors have also been fabricated with other materials and methods [4]–[7].

The memristors are nanoscale devices that store information without need for a power supply, and can be used for nonvolatile memory, and promise to allow imitation of learning, adaptation and spontaneous behavior [8]–[10]. Due to their unique electrical properties, memristors have many potential applications such as nonvolatile memories [11], artificial neural networks [12]–[17], logic/computational circuits [18]–[19], nonlinear circuit designs [4], [20], [22]–[25], and so on.

The most distinguished feature of memristors is the pinched hysteresis loop in the voltage–current plane when driven by any periodic voltage or current signal that has

a zero mean. The pinched hysteresis loops are in fact the hallmarks of all memristors, ideal or otherwise [26]. Because every memristor has a unique set of fingerprints and signatures, the pinched hysteresis loops are used as the fingerprint to identify memristors.

Theoretically, an ideal hysteresis loop must pass through the origin for any bipolar periodic driving signal which has a zero mean [27]. However, experiments have found that some real physical memristors often deviate from the origin over a broad range of amplitude and frequency of the input signal [28]–[32].

Reference [28] points out that the deviation of the pinched point is a parasitic effect, and proposes a generic memristor circuit model by adding four parasitic elements: a capacitor, an inductor, a DC current source and a DC voltage source, to the memristive device.

In 2016, Itoh and Chua comprehensively analyzed the parasitic effects of the genetic memristor in [33]. The paper pointed out that parasitic elements have a significant effect on the dynamics of memristor circuits, and a passive memristor with a strictly-increasing constitutive relation will eventually lose its stored flux when the power is turned off if there is a parasitic capacitor across the memristor. Reference [33] also showed that a parasitic element can be used to break an impasse point, resulting in the emergence of a continuous oscillation in the circuit.

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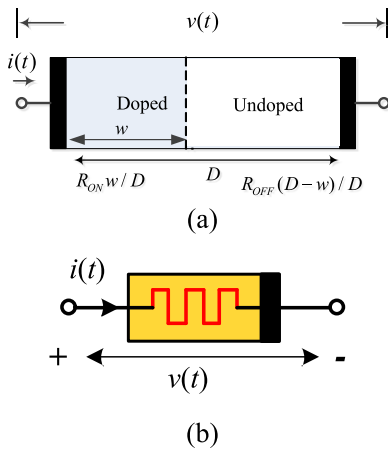


FIGURE 1. HP TiO₂ memristor and its circuit symbol. Note that there are doped region and undoped region in the device and the resistances are $R_{ON}w/D$, and $R_{OFF}(D-w)/D$ respectively.

At present, there is little research on parasitic effects of memristor. The few studies only focus on the general parasitic models of memristors [28], [33] and lack the analysis on parasitic effects of real physical memristors, such as the most famous and practical HP TiO₂ memristor, and NbO₂ Mott memristor [3], [4].

This paper focuses on the parasitic memcapacitor effects of the HP TiO₂ memristor. We assume a parasitic memcapacitor across the HP TiO₂ memristor, and establish the model of the parasitic memcapacitor for analyzing the parasitic effects of the memristor. We show that the pinched point on the hysteresis loop of the memristor does not pass through the origin, and the memristor will eventually lose its stored flux when we switch off its power. To study the parasitic effects in nonlinear memristor based circuits, we design a simple resistor-inductor-memristor based circuit with the parasitic memcapacitor, and find that a parasitic periodic oscillation appears in the circuit.

II. PARASITIC MEMCAPACITOR OF HP MEMRISTOR

Fig. 1 (a) shows the structure of the HP TiO₂ memristor [3], in which a thin titanium dioxide TiO₂ layer and a thin oxygen-deficient titanium dioxide TiO_{2-x} layer are sandwiched between two metal electrodes. When a voltage is applied to the device, the dividing line (i.e., the thickness of the TiO_{2-x} layer) between the TiO₂ and TiO_{2-x} layers shifts with mobility μ_v as a function of the applied voltage. As a result, the total resistance between the two electrodes is altered by two variable resistors connected in series.

The relation between the voltage and current of the HP TiO₂ memristor is described as:

$$\begin{cases} v(t) = \left[\frac{R_{ON}W(t)}{D} + \left(R_{OFF} - \frac{R_{OFF}W(t)}{D} \right) \right] i(t) \\ \frac{dw(t)}{dt} = \mu_v \frac{R_{ON}i(t)}{D} \end{cases} \quad (1)$$

where v and i are the voltage and current across the device respectively; D and w denote the thickness of the sandwiched

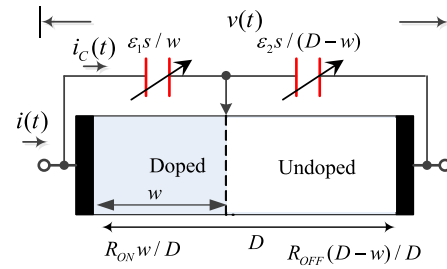


FIGURE 2. Memristor with parasitic memcapacitors.

region and TiO_{2-x} layer of the device respectively; R_{OFF} and R_{ON} denote the high resistance and low resistance for $w = 0$ and $w = D$ respectively.

The microstructure itself of the memristor is a parasitic parallel plate capacitor, and is a memcapacitor that accompanies the change of the memristor, as shown in Fig. 2.

Actually, the parasitic memcapacitor is formed by two parallel-plane structures in series.

The total parasitic memcapacitor can be viewed as two series memcapacitors, which correspond to the doped zone and the undoped zone. The capacitances of the two memcapacitors C_1 and C_2 are respectively described as:

$$c_1 = \frac{\epsilon_1 s}{w} \text{ and } c_2 = \frac{\epsilon_2 s}{D-w} \quad (2)$$

where ϵ_1 and ϵ_2 are the permittivity of doped layer and undoped layer, respectively; s is the cross-sectional area of the memristor.

From equation (2), we can see C_1 and C_2 are indeed memcapacitors. Prove is as follows: For a voltage-controlled generic memcapacitor, the definition function is:

$$\begin{cases} q = C(x)v, \\ \frac{dx}{dt} = g(x, v) \end{cases} \quad (3)$$

Let $x = w$ and $\frac{dw}{dt} = \mu_v \frac{R_{ON}}{D} \frac{v(t)}{[\frac{R_{ON}w(t)}{D} + (R_{OFF} - \frac{R_{OFF}w(t)}{D})]}$, equation (3) can be transformed to:

$$\begin{cases} q = C(w)v, \\ \frac{dw}{dt} = \mu_v \frac{R_{ON}}{D} \frac{v(t)}{[\frac{R_{ON}w(t)}{D} + (R_{OFF} - \frac{R_{OFF}w(t)}{D})]} \end{cases} \quad (4)$$

From equation (3), $C_1 = C_1(w)$ and $C_2 = C_2(w)$ are functions of state variable w , hence, they are memcapacitors.

The total parasitic memcapacitance C is determined by the two variable memcapacitors connected in series:

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{\epsilon_1 \epsilon_2 s / D}{\epsilon_1 + (\epsilon_1 - \epsilon_2) \frac{w}{D}} \quad (5)$$

Two extreme values of the parasitic memcapacitance correspond to two cases, i.e. $C_{\min} = \epsilon_2 s / D$ for $w = 0$ and $C_{\max} = \epsilon_1 s / D$ for $w = D$. Let $\epsilon_1 = 2000 \times 8.85 \times 10^{-12}$ F/m, $\epsilon_2 = 200 \times 8.85 \times 10^{-12}$ F/m, $s = 2 \times 2 \times 10^{-12}$ m² and $D = 10$ nm, we may qualitatively draw the diagram

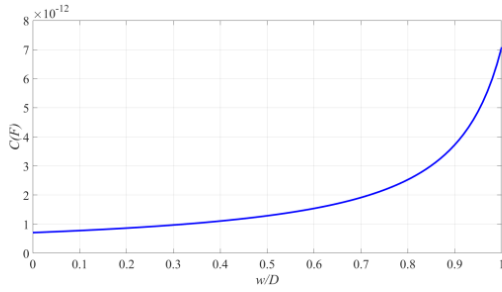


FIGURE 3. Parasitic memcapacitance C versus w/D .

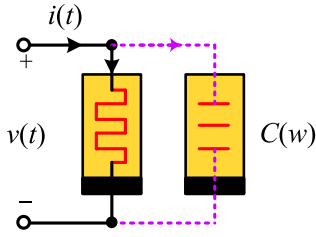


FIGURE 4. Memristor with a parasitic memcapacitor.

of C versus w/D , as shown in Fig. 3. When $w/D \rightarrow 0$, the total memcapacitance tends to $\varepsilon_2 s/D$. While $w/D \rightarrow 1$, the memcapacitance tends to $\varepsilon_1 s/D$. The reason for choosing $\varepsilon_1 = 2000 \times 8.85 \times 10^{-12}$ F/m is that the doped side of the memristor is indeed a conductor and the relative dielectric constant for a conductor tends to infinity.

All Metal-Oxide-Metal (MOM) or Metal-Insulator-Metal (MIM) structure memristors have similar parasitic memcapacitors, so it is very necessary and important to study their mechanisms and influences on memristor performances. By considering the memristor with the parasitic memcapacitor shown in Fig. 4. Note that the memcapacitance is modeled as equation (5).

The governing equation (1) can be transformed into the form:

$$\begin{cases} v(t) = \left[\frac{R_{ON}w(t)}{D} + \left(R_{OFF} - \frac{R_{OFF}w(t)}{D} \right) \right] (i(t) - i_C(t)) \\ \frac{dw(t)}{dt} = \mu_v \frac{R_{ON}(i(t) - i_C(t))}{D} \end{cases} \quad (6)$$

From the second equation of (1), we can deduce the relation between $w(t)$ and $q_M(t)$, where $q_M(t)$ the charge of memristor, shown as follows:

$$\frac{dw(t)}{dt} = \mu_v \frac{R_{ON}}{D} \frac{dq_M(t)}{dt} \quad (7)$$

Integrating both sides of (7), the relationship of $w(t) - q_M(t)$ can be established:

$$w(t) = \mu_v \frac{R_{ON}}{D} q_M(t) \quad (8)$$

Further we notice the bounds of $q_M(t)$, since $0 \leq w(t) \leq D$, and deduce from (6) that:

$$0 \leq q_M(t) \leq \frac{D^2}{\mu_v R_{ON}} \quad (9)$$

From the first equation of (6), we obtain (10):

$$\frac{d\varphi(t)}{dt} = [R_{OFF} - (R_{OFF} - R_{ON})\mu_v \frac{R_{ON}}{D^2} q_M(t)] \frac{dq_M(t)}{dt} \quad (10)$$

Integrating both sides of (10), the expression of $q_M(t)$ can be obtained:

$$q_M(t) = \frac{R_{OFF} \pm \sqrt{R_{OFF}^2 - 2\mu_v \frac{R_{ON}}{D^2} (R_{OFF} - R_{ON})\varphi(t)}}{\mu_v \frac{R_{ON}}{D^2} (R_{OFF} - R_{ON})} \quad (11)$$

Noticing that there are two solutions of $q_M(t)$, however by considering (9), we take the minus sign:

$$q_M(t) = \frac{R_{OFF} - \sqrt{R_{OFF}^2 - 2\mu_v \frac{R_{ON}}{D^2} (R_{OFF} - R_{ON})\varphi(t)}}{\mu_v \frac{R_{ON}}{D^2} (R_{OFF} - R_{ON})} \quad (12)$$

The above equation is messy and it can be further simplified by noting that $R_{OFF} \gg R_{ON}$:

$$q_M(t) = \frac{D^2}{\mu_v R_{ON}} \left(1 - \sqrt{1 - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}} \varphi(t)} \right) \quad (13)$$

Next, we consider C as the parasitic memcapacitor, and our aim is to express C into a function of. According to (5) and by considering $\varepsilon_1 \gg \varepsilon_2$, it can be obtained as:

$$\begin{aligned} C &\approx \frac{\varepsilon_2 s}{D - w(t)} = \frac{\varepsilon_2 s}{D - \mu_v \frac{R_{ON}}{D} q_M(t)} \\ &= \frac{\varepsilon_2 s}{D \sqrt{1 - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}} \varphi(t)}} \end{aligned} \quad (14)$$

If a voltage $v(t) = \sin(2\pi ft)$ is applied to the combination of the memristor and the parasitic memcapacitor, then the equation of $\varphi(t)$ is obtained, which is a time integration of $v(t)$:

$$\varphi(t) = \frac{1}{2\pi f} (1 - \cos(2\pi ft)) \quad (15)$$

Further, the equation of can be obtained:

$$\begin{aligned} i_C(t) &= \frac{dq_C}{dt} = \frac{d(Cv)}{dt} \\ &= \frac{\varepsilon_2 s}{D} \left(\frac{\mu_v \frac{R_{ON}}{D^2 R_{OFF}}}{(1 - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}} \frac{1}{2\pi f} (1 - \cos(2\pi ft)))^{3/2}} \right. \\ &\quad + \frac{(2\pi f - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}}) \cos 2\pi ft}{(1 - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}} \frac{1}{2\pi f} (1 - \cos(2\pi ft)))^{3/2}} \\ &\quad \left. + \frac{\mu_v \frac{R_{ON}}{D^2 R_{OFF}} \cos^2(2\pi ft)}{(1 - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}} \frac{1}{2\pi f} (1 - \cos(2\pi ft)))^{3/2}} \right) \end{aligned} \quad (16)$$

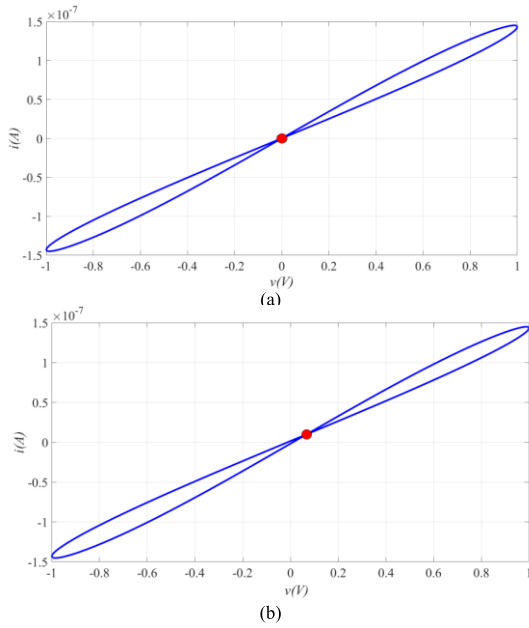


FIGURE 5. Pinched hysteresis loops for a sine voltage input, (a) without parasitic effect, (b) with the parasitic effect caused by a memcapacitor.

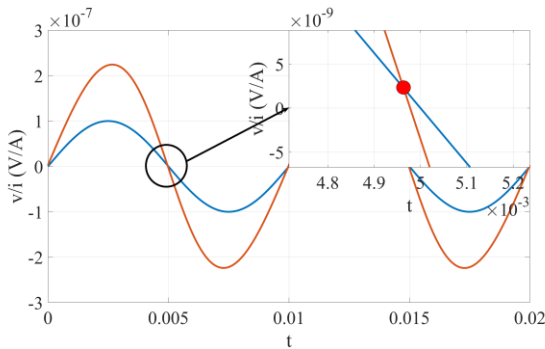


FIGURE 6. Time domain diagram showing the Cross points of voltage and current bias from the x-axis. The voltage is scaled to $1 \times 10^{-7} \sin(2\pi ft)$.

Let $f = 100\text{Hz}$, $\mu_v = 3 \times 10^{-11} \text{m}^2 \text{s}^{-1} \text{V}^{-1}$, $R_{ON} = 1000\Omega$, $R_{OFF} = 5\text{M}\Omega$, $D = 10^{-8} \text{m}$, $\varepsilon_1 = 2000 \times 8.85 \times 10^{-12} \text{F/M}$, $\varepsilon_2 = 200 \times 8.85 \times 10^{-12} \text{F/M}$ and $s = 2 \times 2 \times 10^{-12} \text{m}^2$, we may obtain the new voltage-current relationship diagram, i.e., the pinched hysteresis loop, as shown in Fig. 5 (b). Comparing with the zero-crossing pinched hysteresis loop without parasitic effect, as shown in Fig. 5 (a), we can see that the pinched hysteresis loop may deviate from the origin, showing the parasitic effect to the HP TiO₂ memristor. We can see from Fig. 6 that cross points of voltage and current, which map to the pinched point on the hysteresis loop, bias from the x-axis, which also shows the parasitic effect.

The mechanism of the deviation of the pinched point is as follows:

When considering the parasitic memcapacitor in parallel with the memristor, the phase difference between the voltage

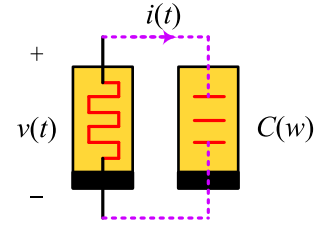


FIGURE 7. Memristor when the power is switched off.

and current of the memristor may be changed significantly, so that the voltage and the current do not pass through time axis simultaneously.

From another point of view, when the driving current begins to be applied to the memristor, $i(t_{0+}) \neq 0$, the voltage across the memcapacitor (or the memristor) cannot be changed abruptly, i.e., $v(t_{0+}) = v(t_{0-}) = 0$; similarly, for any time $t > 0$, when $i(t) = 0$, memristor voltage (or memcapacitor voltage) $v(t) \neq 0$, because there are still a certain amount of static charges on the memcapacitor, which cannot be discharged abruptly.

III. FLUX DECAY VIA THE PARASITIC MEMCAPACITOR

When we switch off the power of the memristor at $t = t_0$, such that terminal current $i = 0$ for $t > t_0$, the circuit shown in Fig. 4 can be changed into Fig. 7.

The governing equations of the circuit can be written as:

$$C \frac{d\varphi}{dt} = -qM(t) \quad (17)$$

According to equation (13) and (14), (17) can be expressed as follows:

$$\frac{d\varphi}{dt} = - \frac{\frac{D^2}{\mu_v R_{ON}} (1 - \sqrt{1 - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}} \varphi(T)})}{\frac{\varepsilon_2 s}{D \sqrt{1 - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}} \varphi(t)}}} \quad (18)$$

By solving equation(17) through MATLAB, the relation between φ and time t can be obtained in Fig. 8 It follows that the HP TiO₂ memristor will eventually lose its initial values $\varphi(t_0)$ via its parasitic memcapacitor (connected in parallel) when we switch off the power of the memristor at $t = t_0$. For the memristor, we can observe that it takes about 10us to release the flux.

IV. PARASITIC OSCILLATION IN A SIMPLE MEMRISTOR CIRCUIT

Let us consider the simple three-element circuit in Fig. 9, where a memristor without parasitic element is in series with an inductor L and a negative resistor $-R$.

From Fig.9, the following differential equations can be obtained:

$$\begin{cases} \frac{dq(t)}{dt} = \frac{R}{L} q(t) \varphi(t) \\ \varphi(t) = R_{OFF}(q(t) - \frac{1}{2} \mu_v \frac{R_{ON}}{D^2} q^2(t)) \end{cases} \quad (19)$$

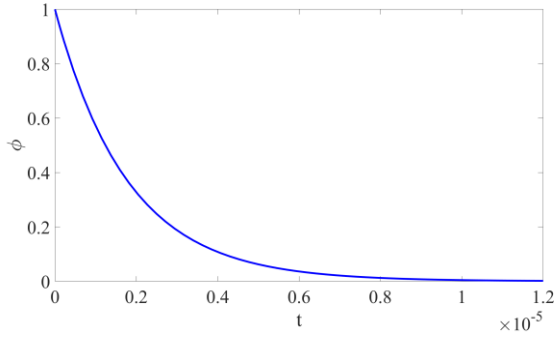


FIGURE 8. Flux and voltage decay of HP TiO₂ memristor.

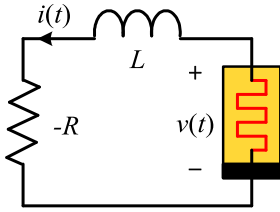


FIGURE 9. A three-element circuit which consists of an active resistor, an inductor, and a memristor.

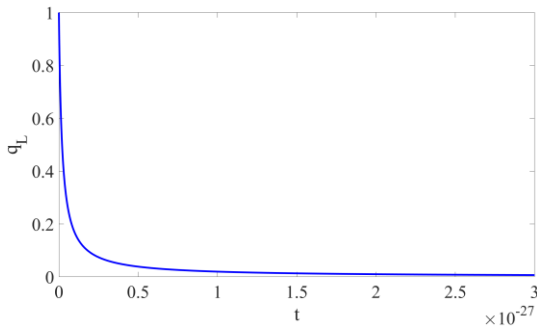


FIGURE 10. Time domain waveform of $q_L - t$ which shows the discharging process.

where $\varphi(t)$ comes from equation (15), solving the equation for $\varphi(t)$, we obtain the second equation of (19).

By solving (19), an analytic solution can be deduced, which is shown below:

$$q(t) = \frac{2D^2}{\mu_1 R_{ON} R_{OFF}} (R + R_{OFF}) \times \left(\frac{\frac{q(0)}{\mu(0) - \frac{2D^2}{\mu_1 R_{ON} R_{OFF}} (R + R_{OFF})} e^{(R + R_{OFF})t}}{\frac{q(0)}{\mu(0) - \frac{2D^2}{\mu_1 R_{ON} R_{OFF}} (R + R_{OFF})} e^{(R + R_{OFF})t} - 1} \right) \quad (20)$$

Fig.10 depicts the solution of (19), where the initial condition is $q(0) = 1$, obviously there is no oscillation. It is just a process of discharging. However, if considering the parasitic memcapacitor of HP TiO₂ memristor, we should observe a periodic oscillation in this circuit, because the parasitic memcapacitor and the inductor formed an LC tank, the energy provided by $-R$ is transferred from L to C and back forth, hence a periodic oscillation is anticipated.

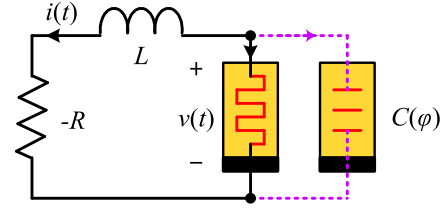
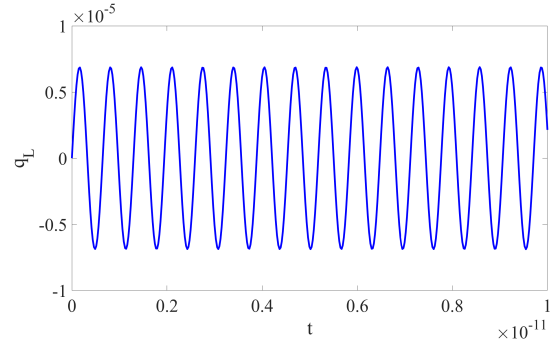
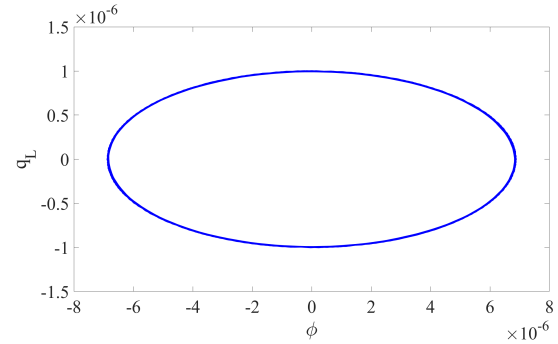


FIGURE 11. Memristor circuit with a parasitic memcapacitance $C(\varphi)$.



(a)



(b)

FIGURE 12. Time domain waveform of q_L and the phase diagram of $q_L - \varphi$, which show periodic oscillation.

In order to study the influence of a parasitic memcapacitor on dynamics of the memristor, a parasitic memcapacitor shown in Fig. 9 is added and Fig.11 is obtained:

The governing equations of the circuit is as follows:

$$\begin{cases} L \frac{dq_L}{dt} = Rq_L + \varphi \\ C \frac{d\varphi}{dt} = -q_L - q_M \end{cases} \quad (21)$$

where q_M and C are given in (13) and (14) respectively. Inserting q_M and C values into (21), we obtain:

$$\begin{cases} \frac{dq_L}{dt} = \frac{R}{L} q_L + \frac{1}{L} \varphi \\ \frac{d\varphi}{dt} = \frac{1}{\frac{\varepsilon_2 S}{D \sqrt{1 - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}}} \varphi(t)}} \times \left[-q_L - \frac{D^2}{\mu_v R_{ON}} \left(1 - \sqrt{1 - 2\mu_v \frac{R_{ON}}{D^2 R_{OFF}}} \varphi(t) \right) \right] \end{cases} \quad (22)$$

Solving (22) by computer, we obtain the time domain waveform of q_L and the phase diagram of $q_L - \varphi$ shown in Fig. 12. Fig. 12(b) shows that there is only one single limit cycle in the phase space, which means that the parasitic memcapacitor may cause periodic oscillation of the original circuit.

V. CONCLUSION

In this paper, a thorough analysis on the parasitic memcapacitor effects of HP TiO₂ memristor is carried out. From the calculation and simulation, we have seen that the parasitic effects lead to the asymmetrical $v(t)$ - $i(t)$ relationship and the flux decay of the memristor, which has potential applications in memristor circuits and systems. Finally, an example memristive oscillator is given to study the parasitic effects in a practical circuit, which shows that the parasitic memcapacitor may cause periodic oscillation.

REFERENCES

- [1] L. Chua, "Memristor-The missing circuit element," *IEEE Trans. Circuit Theory*, vol. CT-18, no. 5, pp. 507–519, Sep. 1971.
- [2] L. O. Chua and S. M. Kang, "Memristive devices and systems," *Proc. IEEE*, vol. 64, no. 2, pp. 209–223, Feb. 1976.
- [3] D. B. Strukov et al., "The missing memristor found," *Nature*, vol. 453, pp. 80–83, May 2008.
- [4] S. Kumar, J. P. Strachan, and R. S. Williams, "Chaotic dynamics in nanoscale NbO₂ Mott memristors for analogue computing," *Nature*, vol. 548, pp. 318–321, Aug. 2017.
- [5] T. D. Dongale et al., "Development of Ag/WO₃/ITO thin film memristor using spray pyrolysis method," *Electron. Mater. Lett.*, vol. 11, no. 6, pp. 944–948, Nov. 2015.
- [6] T. Driscoll, H.-T. Kim, B.-G. Chae, M. Di Ventra, and D. N. Basov, "Phase-transition driven memristive system," *Appl. Phys. Lett.*, vol. 95, no. 4, Jul. 2009, Art. no. 043503.
- [7] F. Corinto and A. Ascoli, "Memristive diode bridge with LCR filter," *Electron. Lett.*, vol. 48, no. 14, pp. 824–825, Jul. 2012.
- [8] M. Di Ventra, Y. V. Pershin, and L. O. Chua, "Circuit elements with memory: Memristors, memcapacitors, and meminductors," *Proc. IEEE*, vol. 97, no. 10, pp. 1717–1724, Oct. 2009.
- [9] H. A. F. Almurib, T. N. Kumar, and F. Lombardi, "Design and evaluation of a memristor-based look-up table for non-volatile field programmable gate arrays," *IET Circuits Devices Syst.*, vol. 10, no. 4, pp. 292–300, Jul. 2016.
- [10] M. Lv and J. Ma, "Multiple modes of electrical activities in a new neuron model under electromagnetic radiation," *Neurocomputing*, vol. 205, pp. 375–381, Sep. 2016.
- [11] M. Elshamy, H. Mostafa, Y. H. Ghallab, and M. S. Said, "A novel non-destructive read/write circuit for memristor-based memory arrays," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 23, no. 11, pp. 2648–2656, Nov. 2015.
- [12] X. Liu, Z. Zeng, and S. Wen, "Implementation of memristive neural network with full-function pavlov associative memory," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 63, no. 9, pp. 1454–1463, Sep. 2016.
- [13] X. Hu, S. Duan, and G. Chen, and L. Chen, "Modeling affections with memristor-based associative memory neural networks," *Neurocomputing*, vol. 223, pp. 129–137, Feb. 2017.
- [14] M. P. Sah, H. Kim, and L. O. Chua, "Brains are made of memristors," *IEEE Circuits Syst. Mag.*, vol. 14, no. 1, pp. 12–36, 4th Quart., 2014.
- [15] S. Wen, T. Huang, Z. Zeng, Y. Chen, and P. Li, "Circuit design and exponential stabilization of memristive neural networks," *Neural Netw.*, vol. 63, pp. 48–56, Mar. 2015.
- [16] M. Prezioso, F. Merrikh-bayat, B. D. Hoskins, G. C. Adam, K. K. Likharev, and D. B. Strukov, "Training and operation of an integrated neuromorphic network based on metal-oxide memristors," *Nature*, vol. 521, pp. 61–64, May 2015.
- [17] X. Han, H. Wu, and B. Fang, "Adaptive exponential synchronization of memristive neural networks with mixed time-varying delays," *Neurocomputing*, vol. 201, pp. 40–50, Aug. 2016.
- [18] I. Vourkas and G. C. Sirakoulis, "Emerging memristor-based logic circuit design approaches: A review," *IEEE Circuits Syst. Mag.*, vol. 16, no. 3, pp. 15–30, 3rd Quart., 2016.
- [19] N. Talati, S. Gupta, P. Mane, and S. Kvaterny, "Logic design within memristive memories using memristor-aided logic (MAGIC)," *IEEE Trans. Nanotechnol.*, vol. 15, no. 4, pp. 635–650, Jul. 2016.
- [20] B. C. Bao, P. Jiang, H. Wu, and F. Hu, "Complex transient dynamics in periodically forced memristive Chua's circuit," *Nonlinear Dyn.*, vol. 79, no. 4, pp. 2333–2343, Mar. 2015.
- [21] Q. Li, H. Zeng, and J. Li, "Hyperchaos in a 4D memristive circuit with infinitely many stable equilibria," *Nonlinear Dyn.*, vol. 79, no. 4, pp. 2295–2308, Mar. 2015.
- [22] B. C. Bao, H. Bao, N. Wang, M. Chen, and Q. Xu, "Hidden extreme multistability in memristive hyperchaotic system," *Chaos, Solitons Fractals*, vol. 94, pp. 102–111, Jan. 2017.
- [23] Q. Lai and S. Chen, "Coexisting attractors generated from a new 4D smooth chaotic system," *Int. J. Control, Automat. Syst.*, vol. 14, no. 4, pp. 1124–1131, Aug. 2016.
- [24] M. Kountchou, P. Louodop, S. Bowong, H. Fotsin, and J. Kurths, "Optimal synchronization of a memristive chaotic circuit," *Int. J. Bifurcation Chaos*, vol. 26, Jun. 2016, Art. no. 1650093.
- [25] Z. I. Mannan, H. Choi, and H. Kim, "Chua corsage memristor oscillator via Hopf bifurcation," *Int. J. Bifurcation Chaos*, vol. 26, Apr. 2016, Art. no. 1630009.
- [26] L. Chua, "If it's pinched it's a memristor," *Semicond. Sci. Technol.*, vol. 29, no. 10, 2014, Art. no. 104001.
- [27] L. Chua, "Everything you wish to know about memristors but are afraid to ask," *Radioengineering*, vol. 24, no. 2, pp. 319–368, Jun. 2015.
- [28] M. P. Sah, C. Yang, H. Kim, B. Muthuswamy, J. Jevtic, and L. Chua, "A generic model of memristors with parasitic components," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 62, no. 3, pp. 891–898, Mar. 2015.
- [29] K. Macvittie and E. Katz, "Electrochemical system with memimpedance properties," *J. Phys. Chem.*, vol. 117, pp. 24943–24947, Nov. 2013.
- [30] A. G. Volkov et al., "Memristors in the venus flytrap," *Plant Signaling Behav.*, vol. 9, May 2014, Art. no. e2920.
- [31] L. Qingjiang, A. Khia, I. Salaoru, C. Papavassiliou, and X. Hui, "Memory impedance in TiO₂ based metal-insulator-metal devices," *Sci. Rep.*, vol. 4, Mar. 2014, Art. no. 4522.
- [32] I. Valov et al., "Nanobatteries in redox-based resistive switches require extension of memristor theory," *Nature Commun.*, vol. 4, Apr. 2013, Art. no. 1771.
- [33] M. Itoh and L. O. Chua, "Parasitic effects on memristor dynamics," *Int. J. Bifurcation Chaos*, vol. 26, 2016, Art. no. 1630014.



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